CV Management System

Streamlining Talent Acquisition with Intelligent CV Processing

MUSTAFA NASSER

Executive Summary

Automating the CV lifecycle to empower recruitment teams.

WHAT IT DOES

Automates the entire CV management lifecycle, from ingestion to intelligent candidate search and outreach.

MAIN BUSINESS VALUE

Reduces manual effort, accelerates recruitment processes, and improves candidate matching precision.

KEY FEATURES

Automated CV ingestion, hybrid candidate search, and integrated email outreach for seamless communication.

High-Level Architecture

Frontend

Streamlit application providing an intuitive user interface for all system interactions.

Backend

FastAPI powers robust and scalable API endpoints for core functionalities.

Storage

File storage for raw CVs and processed data, alongside vector stores and indices.

Interconnections: Frontend communicates with the backend via RESTful APIs, while the backend interacts with storage and indexing layers for data persistence and retrieval.

Core Components

CATEGORY MANAGEMENT

Organizes CVs into logical categories for retrieval.

CV UPLOAD PIPELINE

Ingests and indexes CVs efficiently.

SEARCH ENGINE

Hybrid semantic and keyword candidate search.

EMAIL SERVICE

Automated candidate communication.

VECTOR STORE

Stores numeric representations for semantic search.

UTILITIES

Helper functions and system services.

Category Management

CREATION & INTERFACE

Categories are dynamically defined and managed through the Streamlit frontend, allowing for flexible taxonomy updates.

DISK-BASED STORAGE

Each category maps to a dedicated physical directory, ensuring isolated and organized storage of CVs and metadata.

SYSTEM PROTECTION

Critical system folders are protected via path validation to prevent accidental modification or deletion of core assets.

Purpose: *Enables efficient organization and rapid filtering of candidates based on specific job roles or departmental criteria.*

CV Upload Pipeline

01 **PDF Ingestion**

Upload CV PDF via Streamlit.

02 **Text Extraction**

Extract raw text from the file.

03 **Caching (JSON)**

Save extracted data as {hash}.json.

04 **Profile Generation**

Build structured candidate profile.

05 **Vector Indexing**

Index profile into FAISS/BM25.

PROCESS FLOW

PDF → Text Extraction → Caching → Profile → Indexing

Candidate Profile Generation

Transforming unstructured CV text into standardized, actionable intelligence.

AUTOMATED EXTRACTION

The system parses extracted text to identify and isolate critical candidate data points:

Full Name | Email Address | Technical Skills | Work Experience | Education

LLM-POWERED SUMMARIZATION

Utilizing OpenRouter and GPT-OSS models to synthesize complex career histories into concise professional summaries.

STRATEGIC BENEFITS

Efficiency: Enables rapid screening through standardized profiles.
Searchability: Creates a structured foundation for hybrid semantic and keyword matching.

Caching Strategy

MECHANISM

Stores processed results as `{content_hash}.json` files inside a dedicated `cache/` folder.

CONTENT HASHING

The `content_hash` is computed from the CV's raw text to guarantee uniqueness and avoid duplicate work.

PERFORMANCE BENEFITS

Bypasses extraction & summarization for known files, speeding up listings and previews.

***Efficiency:** Reduces compute and improves response times for frequently accessed CVs.*

Hybrid Search Engine

FAISS (Semantic)

Utilizes vector embeddings to find candidates with conceptually similar skills and experience, capturing intent beyond keywords.

BM25 (Keyword)

Employs traditional probabilistic keyword matching for precise term-based retrieval and exact skill matching.

SCORING WEIGHTS

60% Semantic

40% Keyword

Scores are normalized and combined to provide a comprehensive relevance ranking.

Deduplication: *Results are automatically deduplicated by candidate name and email to ensure a clean, unique list of potential hires.*

Reranking (Optional)

PURPOSE

Improves the relevance and order of search results by applying an additional LLM-based evaluation layer to the initial candidates.

OVER-FETCHING LOGIC

The system initially retrieves a larger pool of candidates (over-fetching) to provide a robust dataset for the LLM to refine.

CONDITIONAL USE

Reranking is an optional feature that can be toggled via configuration to balance between search precision and system performance.

Mechanism: *An LLM re-evaluates the top-N results against the job description to ensure the most qualified candidates appear first.*

Email Subsystem

HTML GENERATION

Leverages LLMs to generate personalized, professional HTML email content tailored to specific candidate profiles and job roles.

SMTP INTEGRATION

Dispatches emails via secure SMTP protocols, with built-in support for Gmail and other major providers.

CANDIDATE RESOLUTION

Automatically resolves candidate contact information from the system's centralized cache for seamless outreach.

***Automation:** Facilitates high-volume, personalized communication with potential candidates, streamlining the recruitment funnel.*

Data Flow: Upload to Indexing

01. PDF UPLOAD
User initiates ingestion by uploading a CV in PDF format.

02. TEXT EXTRACTION
Raw content is extracted and normalized for processing.

03. CONTENT HASHING
Unique SHA-256 hash generated to identify document content.

04. CACHE JSON
Extracted text and metadata persisted to local JSON cache.

05. FAISS/BM25 INDICES
Data is indexed for both semantic and keyword retrieval.

INGESTION LOGIC

The pipeline ensures idempotency through content hashing, preventing duplicate processing while enabling rapid retrieval via a dual-indexing strategy.

Data Flow: Search to Email

01. JOB DESCRIPTION INPUT
User provides requirements for candidate matching.

↓

02. HYBRID SEARCH & RERANK
Combined FAISS/BM25 search followed by optional LLM reranking.

↓

03. RESULTS DISPLAY
Ranked candidates are presented for review and selection.

↓

04. EMAIL GENERATION
LLM creates personalized outreach content for selected candidates.

↓

05. SMTP SEND
Emails are dispatched via the configured SMTP service.

OUTREACH LOGIC

The system bridges the gap between candidate discovery and engagement by automating the generation of highly relevant, personalized communication.

API Endpoints Overview

The backend follows standard RESTful conventions, utilizing FastAPI routers for modularity and clear request/response schemas.

CATEGORIES

- **GET** /categories
- **POST** /categories
- **DELETE** /categories/{id}

CVS

- **POST** /cvs/upload
- **GET** /cvs/{id}
- **DELETE** /cvs/{id}

SEARCH

- **POST** /search/hybrid
- **POST** /search/rerank
- **GET** /search/history

EMAILS

- **POST** /emails/generate
- **POST** /emails/send
- **GET** /emails/status

Streamlit Frontend Pages

DASHBOARD

Provides a high-level overview of system activity, ingestion metrics, and key performance indicators.

UPLOAD PAGE

Centralized interface for users to upload new CVs and initiate the automated processing pipeline.

SEARCH PAGE

Advanced search interface for performing hybrid semantic and keyword-based candidate matching.

PREVIEW PAGE

Displays detailed candidate profiles, extracted skills, and LLM-generated professional summaries.

MANAGE CATEGORIES

Enables administrative control over CV categories, directory structures, and taxonomy organization.

EMAIL TESTER

A utility for validating email generation logic and testing SMTP connectivity for candidate outreach.

Configuration & Environment

EMBEDDING MODEL

Uses `all-MiniLM-L6-v2` for generating vector embeddings, optimized for speed and semantic accuracy.

LLM MODEL

Configurable via OpenRouter; supports models like `GPT-OSS` for summarization and email generation.

SMTP SETTINGS

Requires `SMTP_SERVER`, `SMTP_PORT`, and `SMTP_PASSWORD` for Gmail-integrated outreach.

AUTHENTICATION

Uses `HF_TOKEN` for Hugging Face API access and `OPENROUTER_API_KEY` for LLM services.

FILE PATHS

Configurable directories for `categories/`, `cvs/`, `cache/`, and `indices/` storage.

SEARCH LOGIC

Adjustable `RERANK_FLAG` and `OVERFETCH_LIMIT` to tune search precision vs. latency.

Deployment Considerations

SYSTEM EXECUTION

Backend: `uvicorn main:app --host 0.0.0.0`
Frontend: `streamlit run app.py`

DIRECTORY STRUCTURE

Critical paths: `categories/`, `cvs/`, `cache/`, and `indices/` must be writable by the application user.

MAINTENANCE

Implementation of log rotation and periodic cache validation to ensure long-term system health and disk space management.

Operational Readiness: *Ensure all environment variables are correctly set in the* `.env` *file before launching services.*

Error Handling & Resilience

RESOURCE MONITORING

Proactive disk space checks and file system validation to prevent ingestion failures and ensure storage availability.

GRACEFUL FALLBACKS

System maintains core search functionality even if optional LLM-based reranking services encounter latency or API errors.

DATA INTEGRITY

Idempotent document addition and cache validation logic prevent duplicate entries and maintain a consistent system state.

Resilience: *Designed to handle edge cases in file processing and external API dependencies without compromising user experience.*

Performance & Scalability

IN-MEMORY INDICES

FAISS and BM25 indices are loaded into memory per category, ensuring sub-second search latency even as the document count grows.

OPTIMIZED RERANKING

Strategic over-fetching retrieves a larger candidate pool for LLM reranking, balancing deep evaluation with overall system throughput.

ASYNC PROCESSING

CV uploads and profile generation are handled asynchronously, preventing UI blocking and allowing for parallel ingestion tasks.

***Growth:** Content-based deduplication and modular indexing allow the system to scale horizontally across categories and storage layers.*

Conclusion

SYSTEM STRENGTHS

Fast Hybrid Search

Combines semantic and keyword approaches for superior relevance.

Efficient Caching

Reduces overhead and accelerates retrieval via content hashing.

LLM Integration

Leverages AI for profile summarization and personalized outreach.

NEXT STEPS

Security & Auth

Implement robust user authentication and role-based access control.

Format Expansion

Support additional CV formats beyond PDF, including DOCX.

Cloud Integration

Migrate to cloud storage and serverless compute for scalability.